INTERNAL TEST – I

Course: PCEOT303: Power Systems

Duration: 30 Minutes Maximum Marks: 20

Answer all questions. Each question carries 4 marks

Q1. For a thermal power plant,

a) Arrange the following components in the correct sequence of operation:
Boiler, Turbine, Generator, Condenser

b) Write the primary function of each of the above components.

Q2. The time duration and load of a power plant are given below:

Time (hours)	Load (MW)
0–6	30
6–12	60
12–18	90
18–24	60

a) Draw the load curve

b) Calculate:

i. Daily energy generated, ii. Demand factor, iii. Load factor

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Q3. For a power plant:

Installed Capacity = 120 MW, Connected Load = 160 MW, Maximum Demand = 100 MW, Energy Generated = 1920 MWh/day, Operating Hours = 20 h

Calculate:

a. Demand Factor, b. Plant Capacity Factor, c. Plant Use Factor, d. Average Load

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Q4. Answer the following:

a. Define Load Duration Curve.
b. State one difference between a Load Curve and a Load Duration Curve.
c. State one advantage of a high Diversity Factor.
d. Name the thermodynamic cycle used in a steam power plant.

Q5. Draw the block diagram for a wind power plant
